Predicting Airfoil Self-Noise Using Deep Neural Networks

Analytical Deep Learning Project Report

Task: Supervised learning regression

Dataset: NASA Airfoil Self-Noise — UCI Machine Learning Repository

Framework: TensorFlow / Keras

Final model: Feed-forward neural network (5 → 64 → 32 → 16 → 1)

Final selected architecture: 5 → 64 → 32 → 16 → 1

Final test performance: MAE 3.3268 dB · RMSE 4.3036 dB · R^2 0.6303

1. Project Objective

The objective of this project is to develop a deep neural network capable of predicting airfoil self-noise from aerodynamic measurements. The problem is treated as a supervised regression task because the target is a continuous sound-pressure measurement.

The work also examines how preprocessing, network capacity and learning-rate choices affect generalization. The emphasis is therefore not only on obtaining a prediction, but on comparing controlled model variations and selecting the strongest model using an untouched test set.

2. Dataset Description

The project uses the NASA Airfoil Self-Noise dataset from the UCI Machine Learning Repository. The dataset contains 1,503 observations, five input features and one continuous target.

Variable	Role	Description
frequency_hz	Input	Acoustic measurement frequency (Hz)
angle_of_attack_deg	Input	Angle of attack (degrees)
chord_length_m	Input	Airfoil chord length (m)
free_stream_velocity_ms	Input	Free-stream velocity (m/s)
suction_side_displacement_thickness_m	Input	Suction-side displacement thickness (m)
scaled_sound_pressure_db	Target	Scaled sound pressure level (dB)

Data quality checks found 1,503 non-null values in each column and no missing values. The dataset therefore required no missing-value imputation.

3. Exploratory Data Analysis

The exploratory stage examined numerical summaries, distributions, scatterplots and correlations. Frequency and suction-side displacement thickness showed pronounced right-skew. The other inputs also showed structured experimental distributions rather than simple continuous bell-shaped forms.

The Pearson correlation between frequency and the target was approximately -0.391, making it the strongest individual linear association with scaled sound pressure among the five inputs. Suction-side displacement thickness had a correlation of approximately -0.313 with the target.

Angle of attack and suction-side displacement thickness showed a relatively strong feature-feature correlation of approximately 0.753. This was not used as a reason to remove either variable because the two variables describe different physical quantities and correlation alone does not establish that one is redundant for nonlinear prediction.

Scatterplots did not show simple straight-line relationships between the individual inputs and the target. The structured and nonlinear-looking patterns provided a reasonable basis for testing a multilayer neural network.

4. Data Preprocessing

The five input features were separated from the target before modelling. The data were split into 80% training data (1,202 observations) and 20% testing data (301 observations), using a fixed random state of 42.

StandardScaler was fitted using the training features only. The fitted scaler was then applied to both the training and test features. This produced approximately zero means and unit standard deviations for the scaled training features while preventing test-set information from influencing the scaling parameters.

Frequency transformation experiment

Because frequency was strongly right-skewed, a log1p transformation was tested as a controlled preprocessing experiment. The transformed-frequency model performed worse on the held-out test set, so the original frequency representation was retained.

5. Neural Network Methodology

The selected modelling approach is a feed-forward neural network (multilayer perceptron). The hidden layers use ReLU activation to introduce nonlinear behaviour, while the output layer uses a linear activation because the target is a continuous regression value.

Mean Squared Error (MSE) was used as the training loss. Mean Absolute Error (MAE) was monitored during training and also reported as a final evaluation metric. Adam was used as the optimizer.

Setting	Value
Epochs	100
Batch size	32
Validation split	0.20 of training data
Optimizer	Adam
Baseline learning rate	0.001
Loss	Mean Squared Error
Training metric	Mean Absolute Error

6. Model Experiments

Baseline model

The baseline architecture contained 32 and 16 neurons in its two hidden layers. It contained 737 trainable parameters. Architecture: $5 \rightarrow 32 \rightarrow 16 \rightarrow 1$.

Variation 1 — log-transformed frequency

The baseline architecture was retained while frequency was replaced by log1p(frequency). This isolated the effect of the preprocessing change.

Variation 2 — larger network

The network capacity was increased to 64, 32 and 16 neurons in the hidden layers. The resulting model contained 3,009 trainable parameters. The original feature representation was retained.

Variation 3 — lower learning rate

The Variation 2 architecture was retained, but the Adam learning rate was reduced from 0.001 to 0.0005.

7. Experimental Results

All models were evaluated using the same untouched test set. MAE and RMSE are lower-is-better metrics, while R^2 is higher-is-better.

Experiment	MAE (dB)	RMSE (dB)	R^2
Baseline	4.1206	5.2105	0.4581
Variation 1 — Log Frequency	4.8433	6.1057	0.2559
Variation 2 — Larger Network	3.3268	4.3036	0.6303
Variation 3 — Lower Learning Rate	4.2948	5.4331	0.4108

Variation 2 produced the strongest test performance across all three metrics. Its MAE of 3.3268 dB was the lowest, its RMSE of 4.3036 dB was the lowest, and its R^2 of 0.6303 was the highest.

Compared with the baseline, the larger network reduced MAE from 4.1206 to 3.3268 dB and increased R^2 from 0.4581 to 0.6303. The log-frequency experiment instead reduced performance, while the lower learning rate also underperformed the larger network.

8. Learning Behaviour

The recorded training histories showed substantial improvement during early training followed by slower improvement. Variation 2 showed a relatively stable convergence pattern, with training and validation losses remaining reasonably close during the later stages.

For Variation 3, validation MAE decreased from 124.7921 at epoch 1 to 4.5570 at epoch 100. The final test performance, however, remained below that of Variation 2.

9. Final Model Selection

Variation 2 was selected as the final model because it achieved the best held-out test performance among the tested configurations. The final architecture is 5 → 64 → 32 → 16 → 1, using ReLU in the hidden layers and a linear output.

Final selected architecture: 5 → 64 → 32 → 16 → 1

Final model property	Selected value
Architecture	5 → 64 → 32 → 16 → 1
Hidden activation	ReLU
Output activation	Linear
Optimizer	Adam
Learning rate	0.001
Loss	MSE
Trainable parameters	3,009
Test MAE	3.3268 dB
Test RMSE	4.3036 dB
Test R^2	0.6303

10. Key Findings

Increasing network capacity was the most successful tested intervention. The larger network captured the dataset relationships more effectively than the baseline under the tested settings. The log transformation of frequency, despite reducing visible skew, did not improve predictive performance. Reducing the learning rate from 0.001 to 0.0005 also did not improve generalization.

11. Limitations

The dataset contains 1,503 observations, so the model was trained on a relatively small dataset for a deep-learning study. The experiments used a single train-test split rather than cross-validation. Only a limited number of architectures and hyperparameter settings were tested. The model is a data-driven predictor and should not be interpreted as a complete physical model of airfoil aeroacoustics.

12. Future Work

Future work could use cross-validation, systematic hyperparameter optimization, early stopping, dropout or L2 regularization, alternative optimizers, and broader comparisons with conventional regression and tree-based methods. Further analysis could examine prediction error across frequency, angle of attack, velocity and other

operating regimes.

13. Conclusion

This project demonstrated a complete supervised deep-learning workflow for predicting airfoil self-noise. The work progressed from data inspection and preprocessing to controlled neural-network experiments and held-out evaluation. Among the tested models, the larger $5 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 1$ network provided the strongest performance, achieving an MAE of 3.3268 dB, RMSE of 4.3036 dB and R^2 of 0.6303 on the test set.

14. Reference

Brooks, T. F., Pope, D. S., & Marcolini, M. A. (1989). *Airfoil Self-Noise*. UCI Machine Learning Repository. DOI: 10.24432/C5VW2C.

Dataset source: https://archive.ics.uci.edu/dataset/291/airfoil%2Bself-noise